

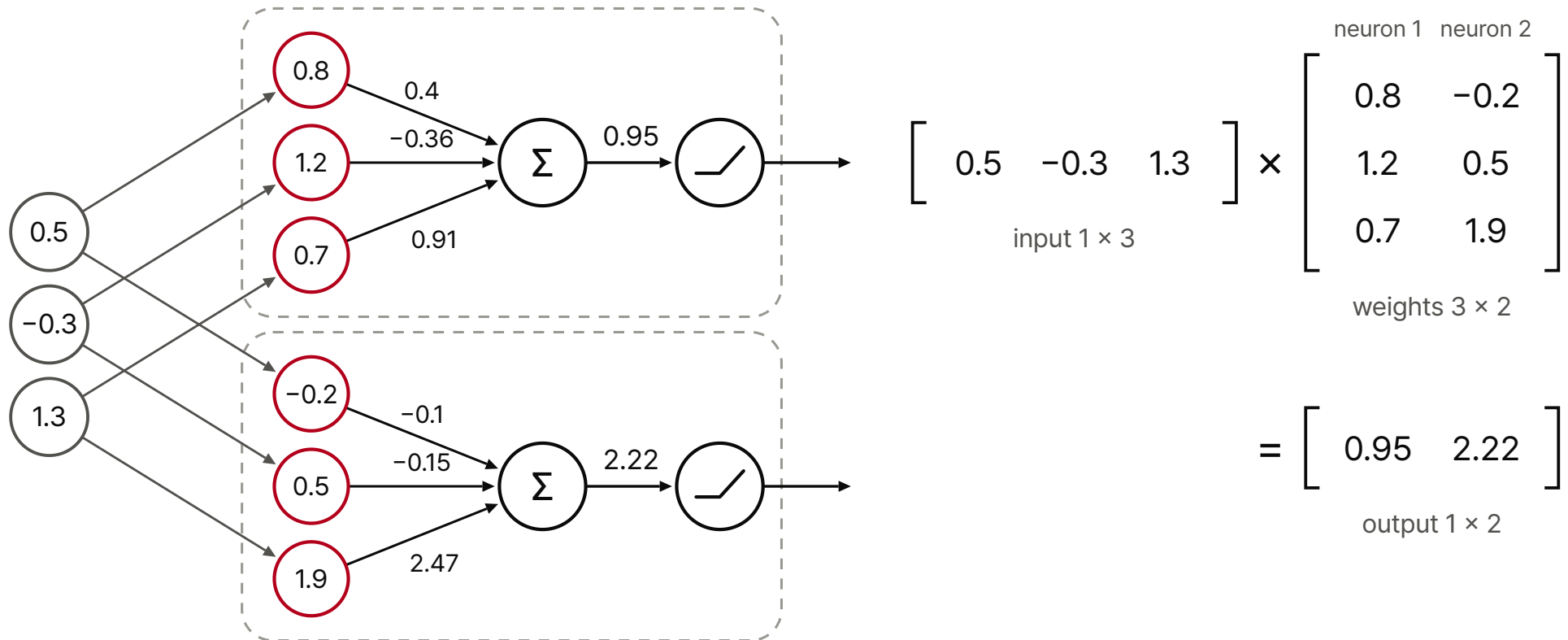
Data & AI

6. Transformations and Backpropagation

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Last Time

One layer is a matrix multiplication



- Each column of W holds one neuron's weights
- Run trains every weight and every bias with the rule from lecture 4

Today

jdasam.github.io/mas1004/demos/lecture06-one-layer-2d · [lecture06-backprop-by-hand](#)

- What a matrix does to the plane, and what the bias and the activation add
- How a trained network moves points, layer by layer
- Backpropagation: every slope from one pass backward through the loss

What a matrix does to the plane

A Point as Input

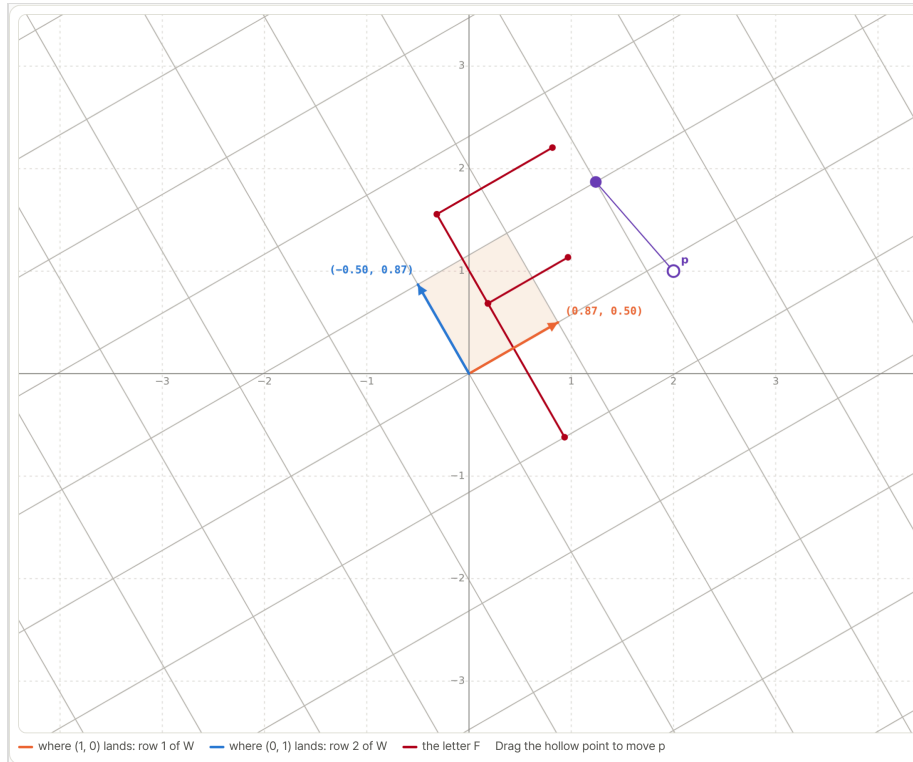
A layer of two neurons, two inputs

$$\begin{bmatrix} x & y \end{bmatrix} \times \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} x \cdot w_{11} + y \cdot w_{21} & x \cdot w_{12} + y \cdot w_{22} \end{bmatrix} = \begin{bmatrix} h_1 & h_2 \end{bmatrix}$$

- The input is a point $p = (x, y)$, and the output is another point (h_1, h_2)
- Same rule as lecture 5: column 1 is neuron 1, column 2 is neuron 2

Where (1, 0) and (0, 1) Go

Demo, preset "Rotate 30°"



W: two neurons, one column each

	N1	N2
(1, 0) →	0.87	0.50
(0, 1) →	-0.50	0.87

b 0.00 0.00

act Linear **ReLU** tanh

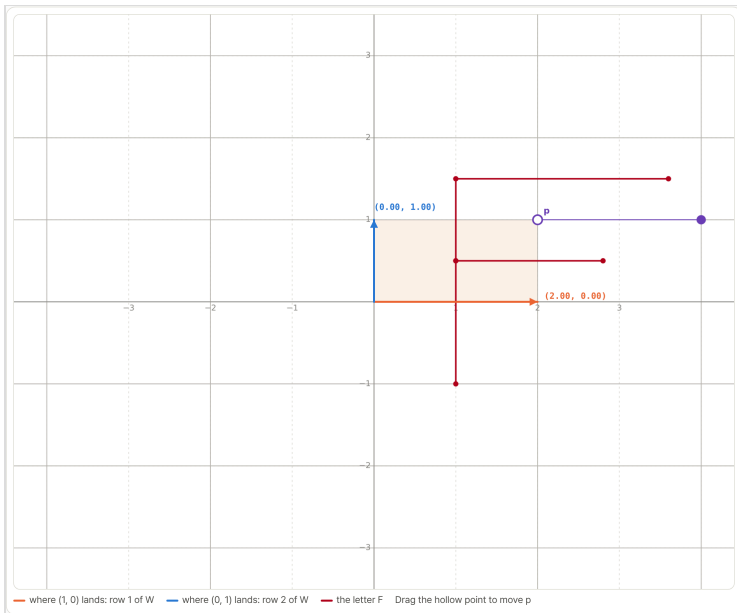
p = (2.00, 1.00)

$$\begin{aligned} & [2.00 \ 1.00] \times \begin{bmatrix} 0.87 & 0.50 \\ -0.50 & 0.87 \end{bmatrix} \\ &= [2.00 \times 0.87 + 1.00 \times (-0.50) \quad 2.00 \times 0.50 + 1.00 \times 0.87] \\ &= [1.24 \ 1.87] \end{aligned}$$

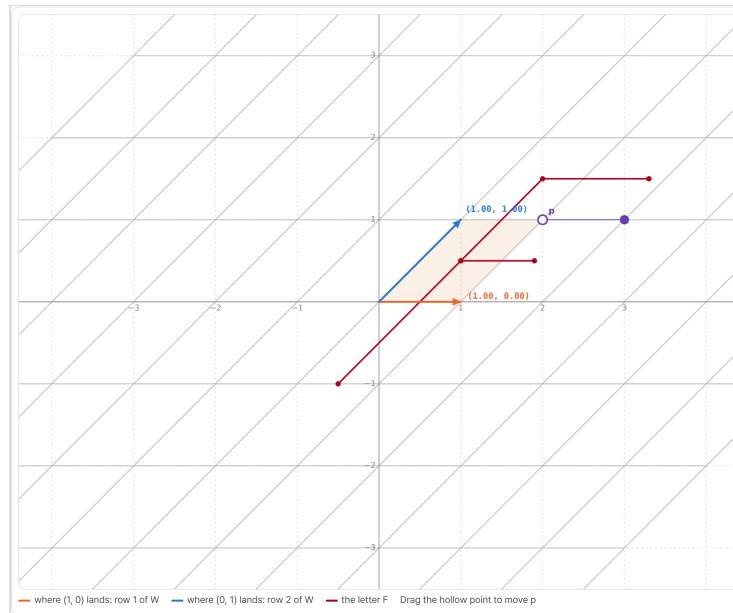
- $(1, 0) \times W$ is the first row of W, and $(0, 1) \times W$ is the second row
- The two arrows tell where every other point goes

Three More Presets

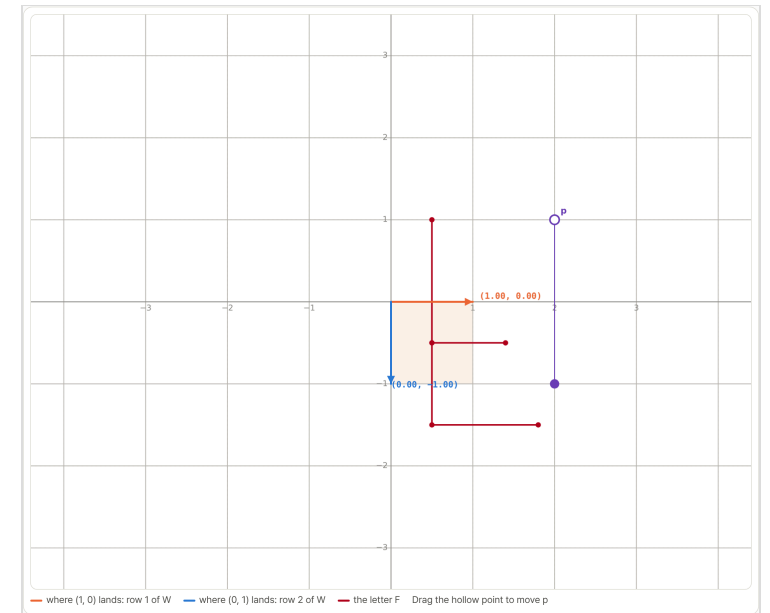
Demo, presets "Stretch", "Shear", "Flip"



Stretch: $W = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$



Shear: $W = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$



Flip: $W = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

- Grid lines stay straight and parallel, and the origin stays put

Matrix Multiplication as a Transformation

3Blue1Brown, Essence of Linear Algebra, Chapter 3

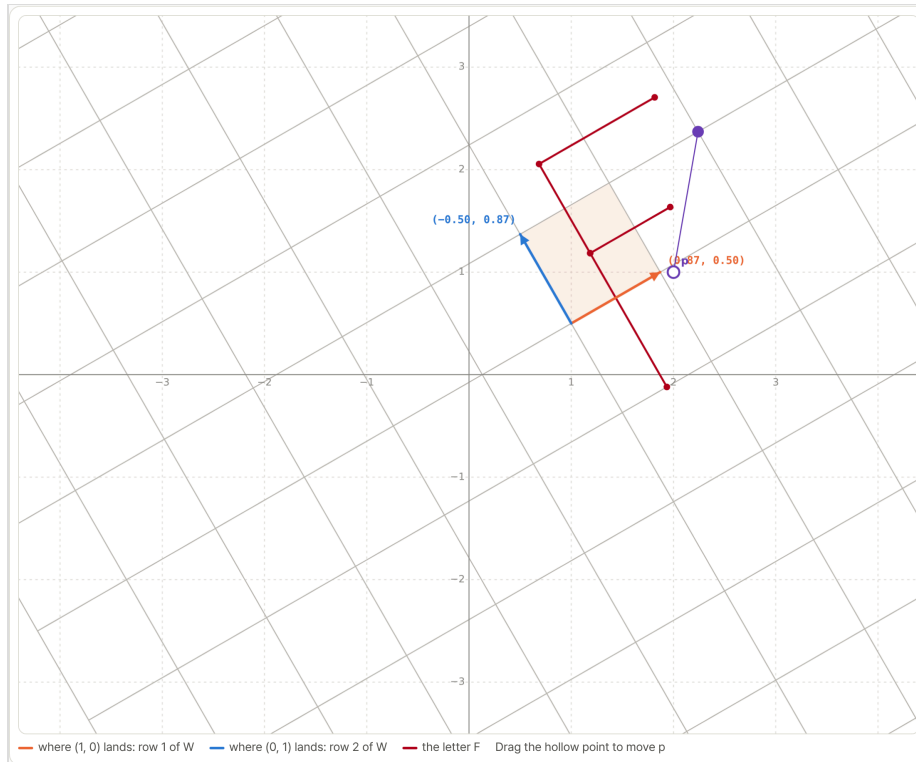
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

Where all the intuition is

- A matrix moves every point of the plane at once: rotate, stretch, shear

Adding the Bias

Demo, "+ b" on



W: two neurons, one column each

	N1	N2
(1, 0) →	0.87	0.50
(0, 1) →	-0.50	0.87

b 1.00 0.50

act Linear ReLU tanh

p = (2.00, 1.00)

$[2.00 \ 1.00] \times \begin{bmatrix} 0.87 & 0.50 \\ -0.50 & 0.87 \end{bmatrix}$

$= [2.00 \times 0.87 + 1.00 \times (-0.50)$

$2.00 \times 0.50 + 1.00 \times 0.87]$

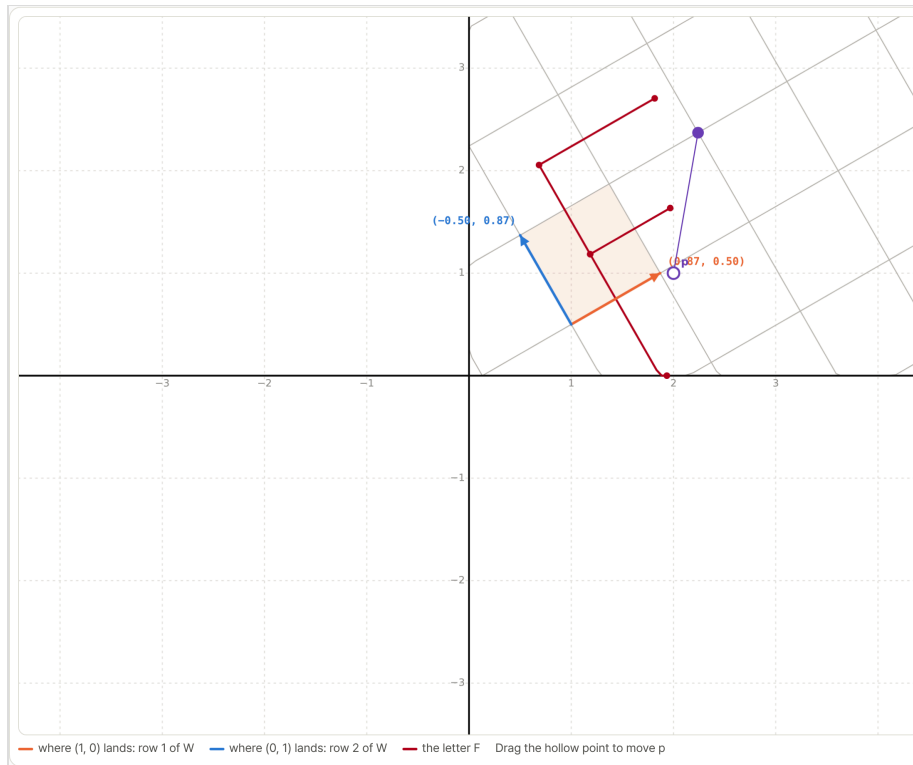
$= [1.24 \ 1.87]$

$+ [1.00 \ 0.50] = [2.24 \ 2.37]$

- The whole plane moves by (b_1, b_2) , and the origin moves with it

ReLU on the Plane

Demo, "act" on, ReLU



W: two neurons, one column each

	N1	N2
(1, 0) →	0.87	0.50
(0, 1) →	-0.50	0.87

b 1.00 0.50

act Linear **ReLU** tanh

p = (2.00, 1.00)

$[2.00 \ 1.00] \times \begin{bmatrix} 0.87 & 0.50 \\ -0.50 & 0.87 \end{bmatrix}$

$= [2.00 \times 0.87 + 1.00 \times (-0.50)$

$2.00 \times 0.50 + 1.00 \times 0.87]$

$= [1.24 \ 1.87]$

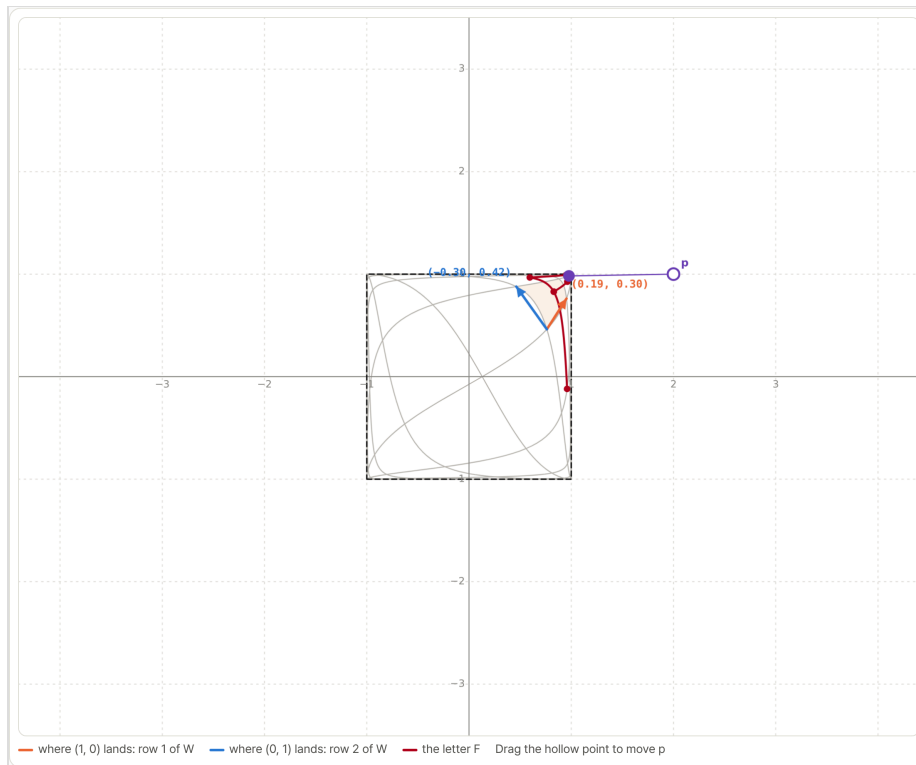
$+ [1.00 \ 0.50] = [2.24 \ 2.37]$

ReLU → [2.24 2.37]

- ReLU sends every negative coordinate to 0, which folds the plane: the first step that bends a straight line

tanh on the Plane

Demo, "act" on, tanh



W: two neurons, one column each

	N1	N2	
(1, 0) →	0.87	0.50	
(0, 1) →	-0.50	0.87	
b	1.00	0.50	
act	Linear	ReLU	tanh

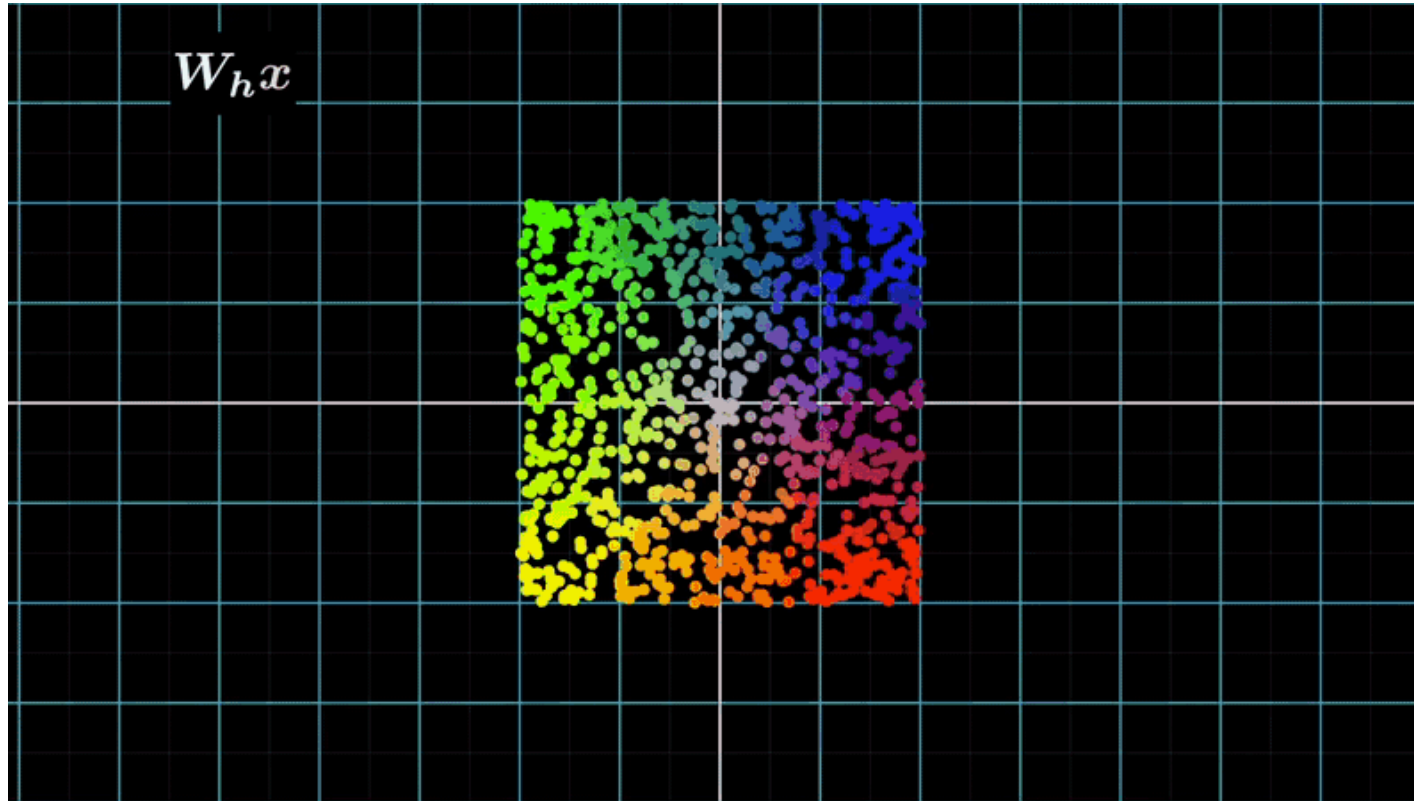
p = (2.00, 1.00)

$$\begin{aligned} & [2.00 \ 1.00] \times \begin{bmatrix} 0.87 & 0.50 \\ -0.50 & 0.87 \end{bmatrix} \\ &= [2.00 \times 0.87 + 1.00 \times (-0.50) \quad 2.00 \times 0.50 + 1.00 \times 0.87] \\ &= [1.24 \ 1.87] \\ &+ [1.00 \ 0.50] = [2.24 \ 2.37] \\ &\text{tanh} \rightarrow [0.98 \ 0.98] \end{aligned}$$

- Every coordinate is pressed into $(-1, 1)$, more strongly the farther it is from 0

ReLU Applied to the Plane

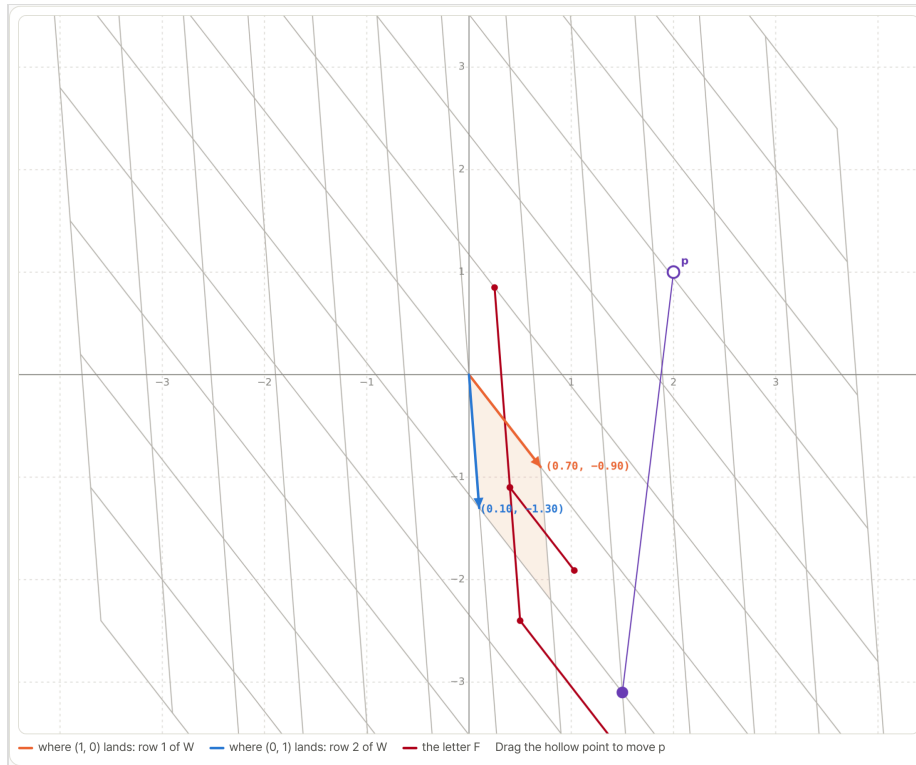
vcubingx, The Neural Network, A Visual Introduction



- $h = (Wx + b)^+$: multiply by W , add b , then ReLU (the $^+$)

A Random Matrix

Demo, preset "Random"



W: two neurons, one column each

	N1	N2
(1, 0) →	0.70	-0.90
(0, 1) →	0.10	-1.30

b

act

p = (2.00, 1.00)

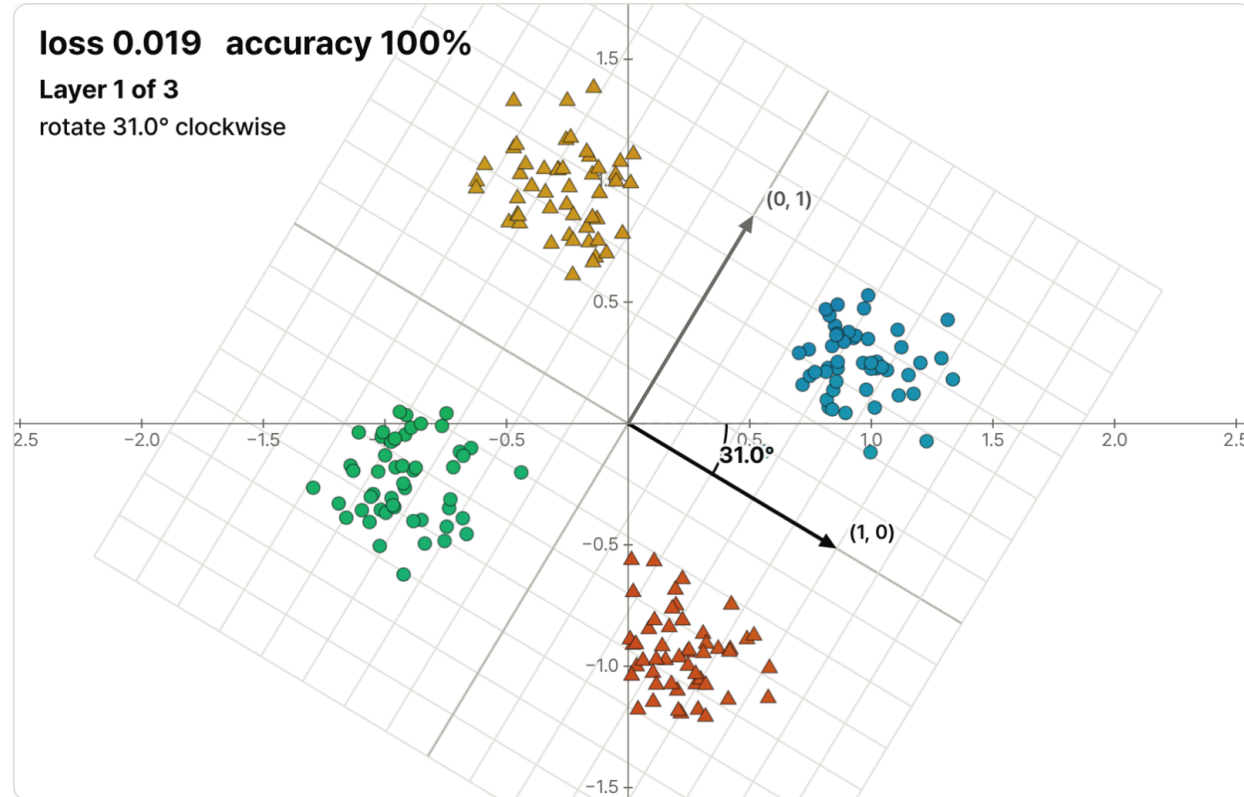
$$\begin{aligned} & [2.00 \ 1.00] \times \begin{bmatrix} 0.70 & -0.90 \\ 0.10 & -1.30 \end{bmatrix} \\ &= [2.00 \times 0.70 + 1.00 \times 0.10 \quad 2.00 \times (-0.90) + 1.00 \times (-1.30)] \\ &= [1.50 \ -3.10] \end{aligned}$$

- Read the two rows of W and say where (1, 0) and (0, 1) go before you press Play
- Then turn on ReLU and say which part of the F gets folded

A trained network, layer by layer

Layer 1 of a Trained Network

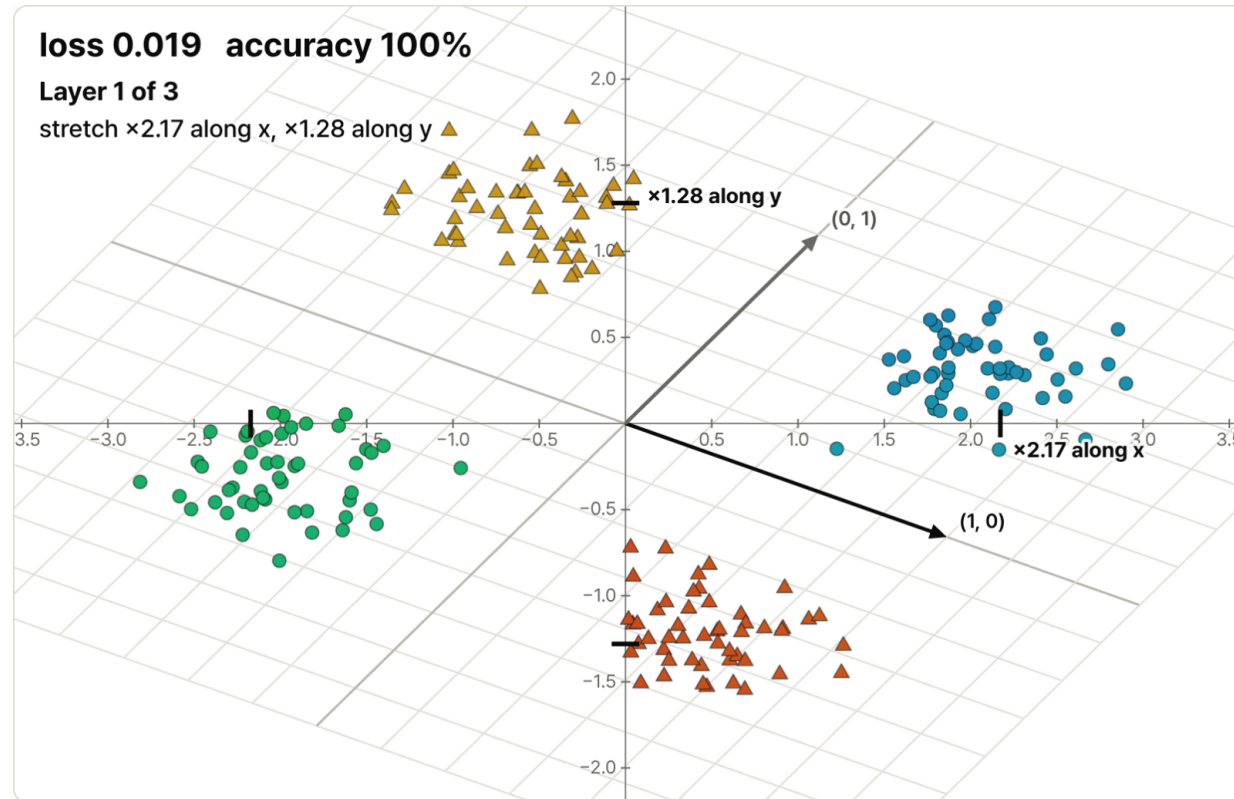
Layer by Layer in 2D · XOR · rotate 31.0° clockwise



- The same steps as the demo, with W and b found by gradient descent

Layer 1 of a Trained Network

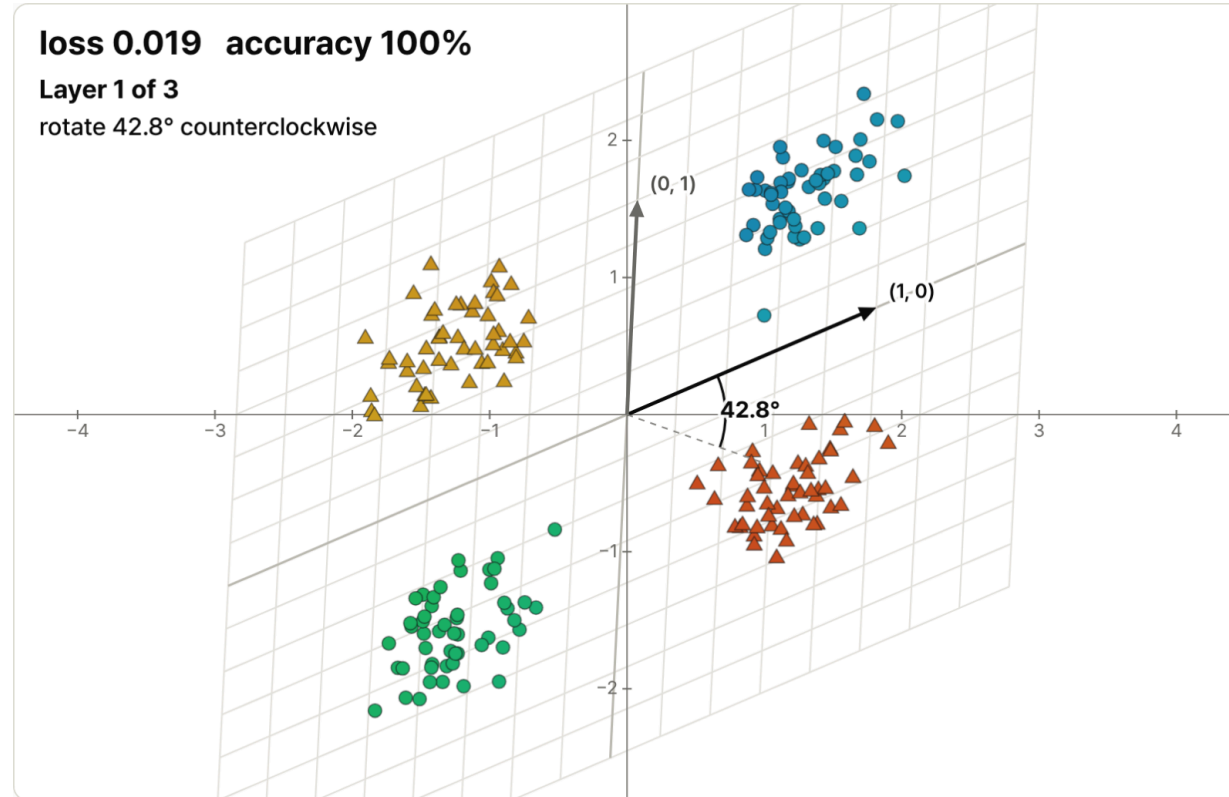
Layer by Layer in 2D · XOR · stretch $\times 2.17$ along x, $\times 1.28$ along y



- The same steps as the demo, with W and b found by gradient descent

Layer 1 of a Trained Network

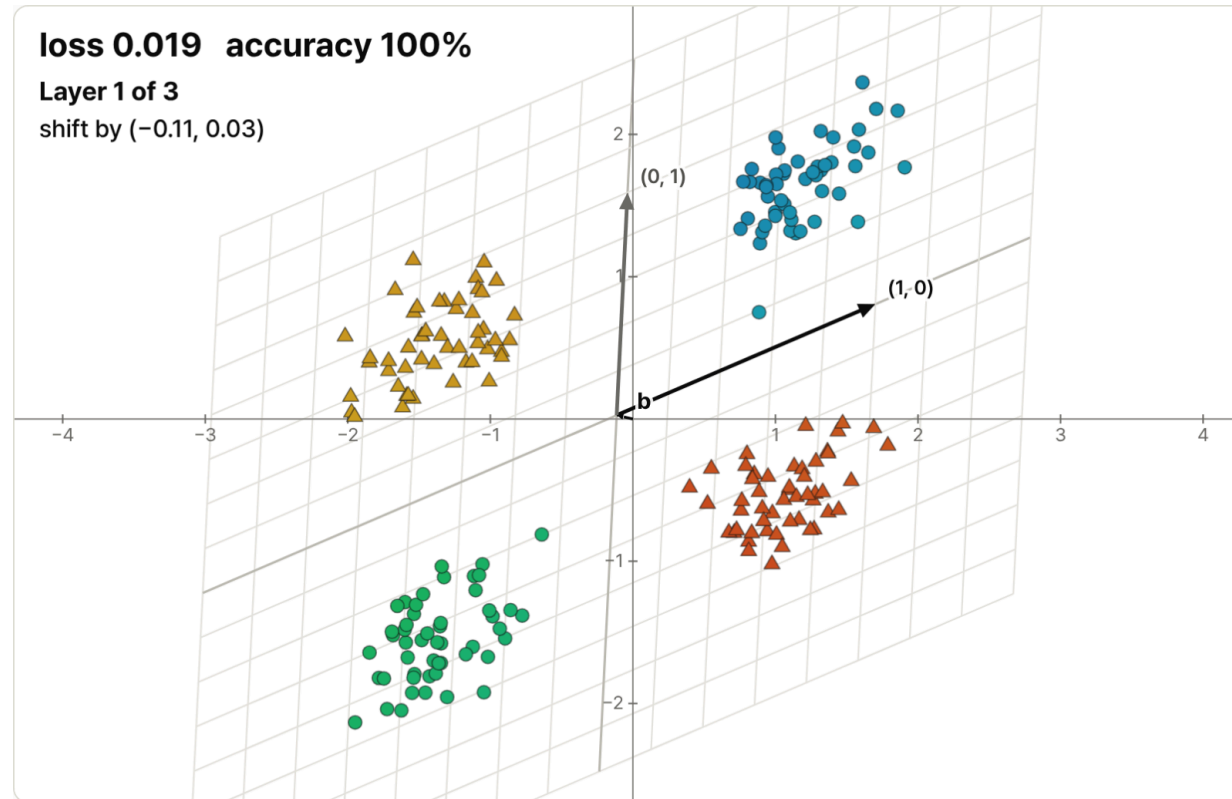
Layer by Layer in 2D · XOR · rotate 42.8° counterclockwise



- The same steps as the demo, with W and b found by gradient descent

Layer 1 of a Trained Network

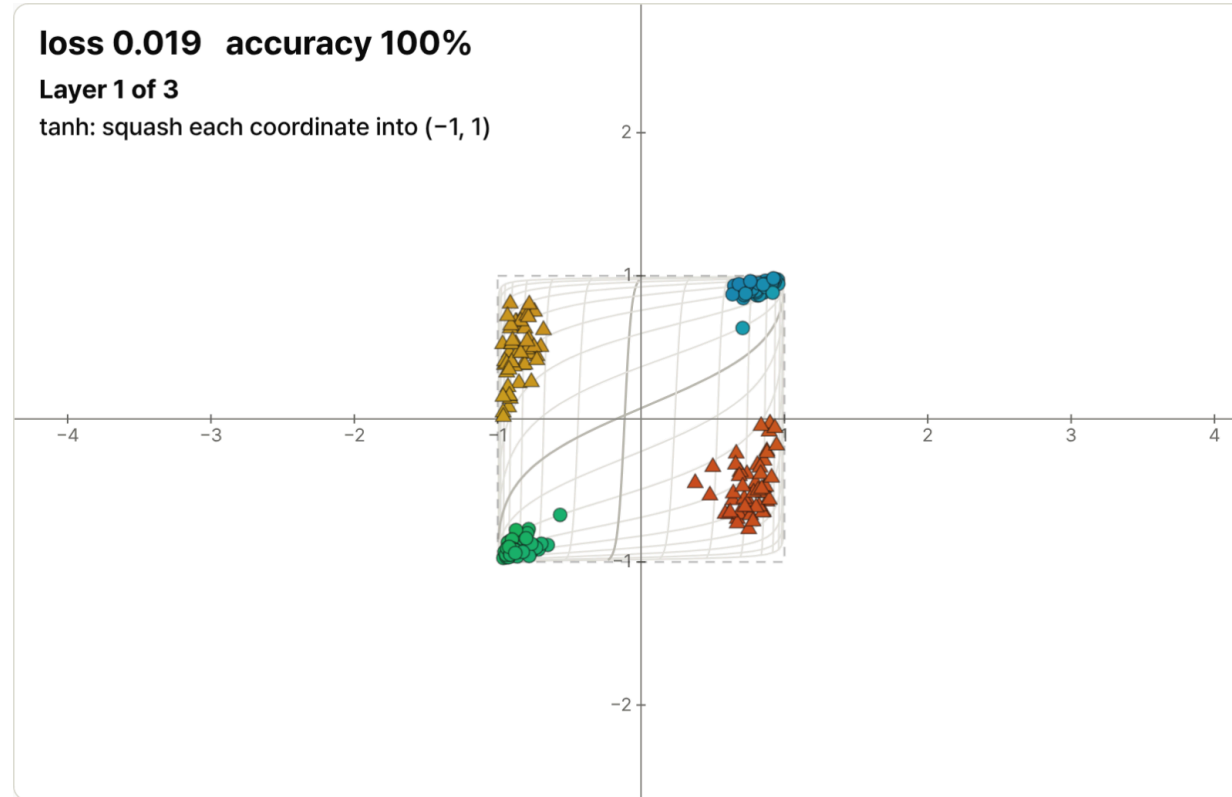
Layer by Layer in 2D · XOR · shift by $(-0.11, 0.03)$



- The same steps as the demo, with W and b found by gradient descent

Layer 1 of a Trained Network

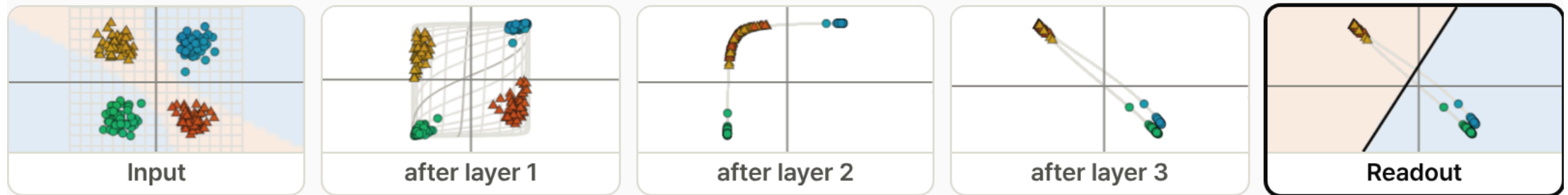
Layer by Layer in 2D · XOR · tanh: squash into $(-1, 1)$



- The same steps as the demo, with W and b found by gradient descent

The Plane After Every Layer

Layer by Layer in 2D · thumbnail strip



- Each layer rotates, stretches, shifts and squashes the plane again
- In the last space one straight line separates the two colors

Crumpled Paper

François Chollet, Deep Learning with Python (2017)



- Two sheets of paper crumpled into one ball
- Layer by layer, the network uncrumples the ball until the sheets come apart

Backpropagation

Measuring Slopes in Lecture 4

Nudge one knob by Δ , compute the loss again

Knobs	Loss computations per step
8 (lecture 4, eight bumps)	9
33 (Neuron Patch, "2 layers")	34
10^{11} (a large language model)	about 10^{11}

- One computation for the current loss, one more for every nudged knob

The Loss as a Chain of Small Operations

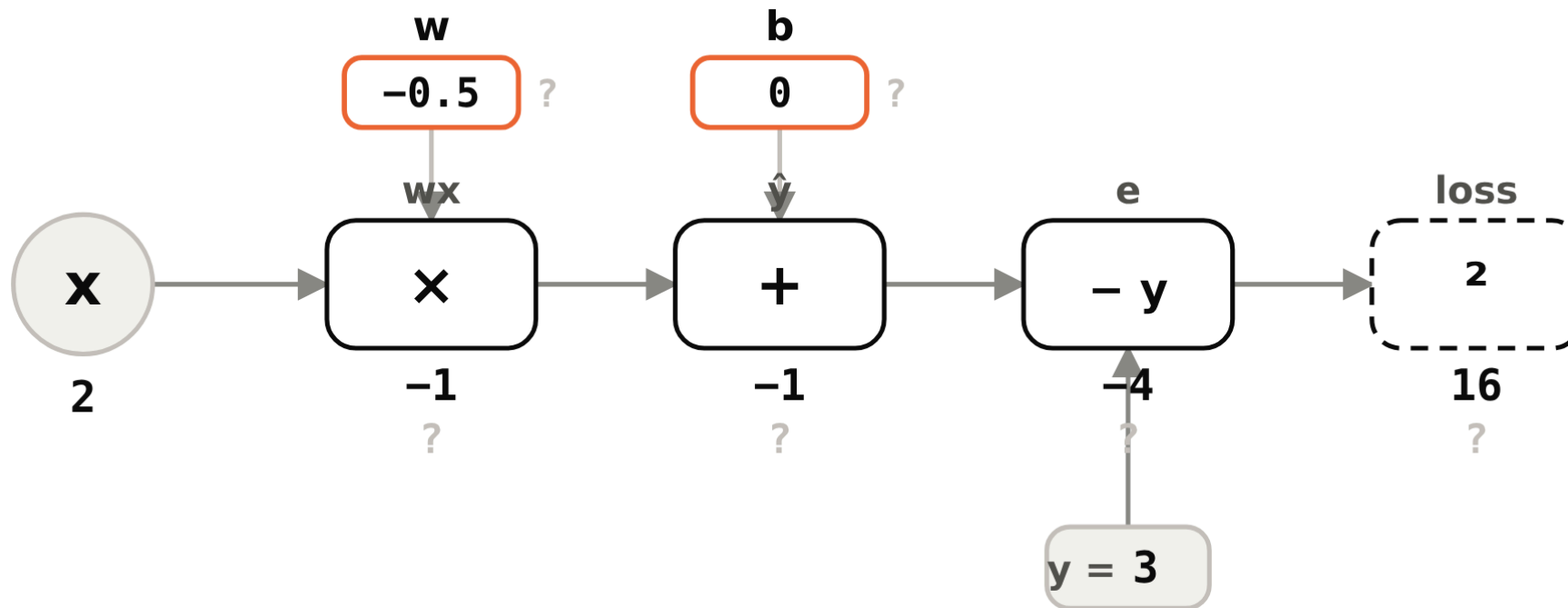
Backprop by Hand, preset "1 neuron", point $(x, y) = (2, 3)$

$$x \rightarrow \times w \rightarrow + b \rightarrow \hat{y} \rightarrow - y \rightarrow e \rightarrow ^2 \rightarrow \text{loss}$$

- Four operations: multiply, add, subtract y , square
- Loss for one point: $(\hat{y} - y)^2$

Forward Pass

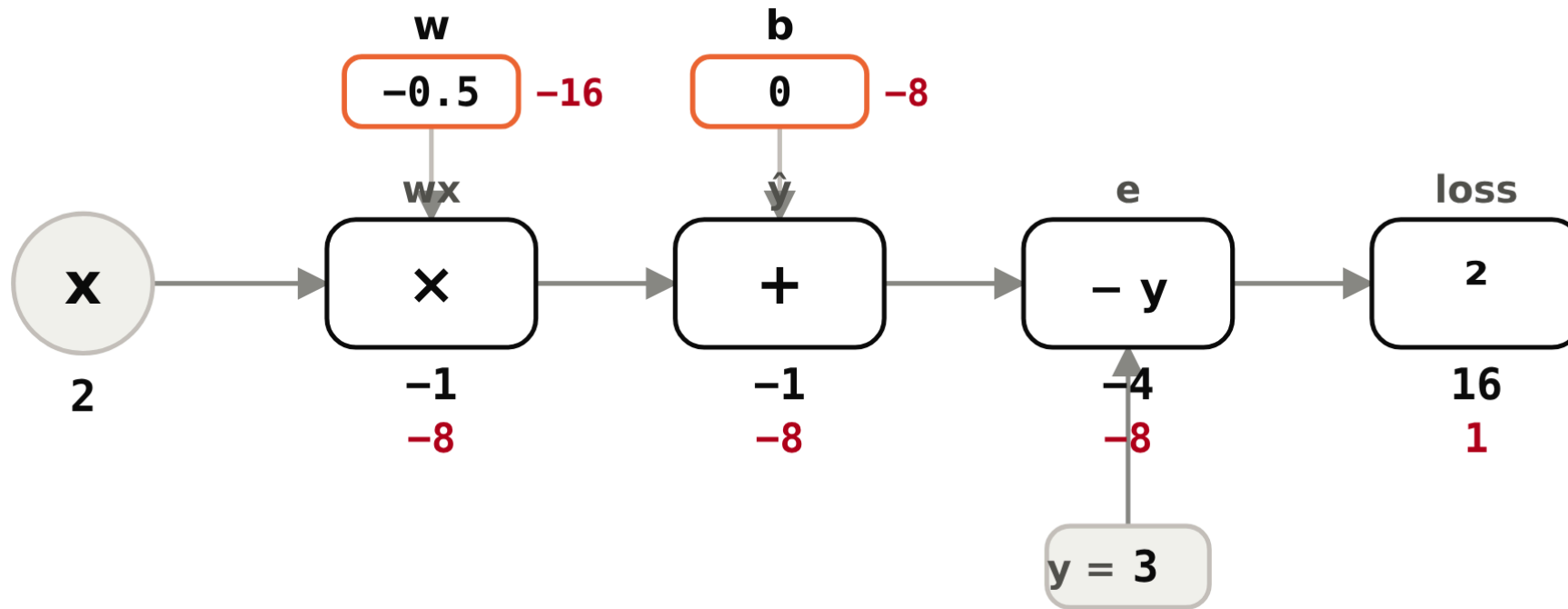
Backprop by Hand, preset "1 neuron"



- $w \cdot x = -1$, $\hat{y} = -1$, $e = \hat{y} - y = -4$, $\text{loss} = 16$

Backward Pass

Backprop by Hand, preset "1 neuron"



- Start at the loss with slope 1, then move one box to the left at a time
- w gets -16 : nudge w by 0.001 and the loss changes by about -0.016

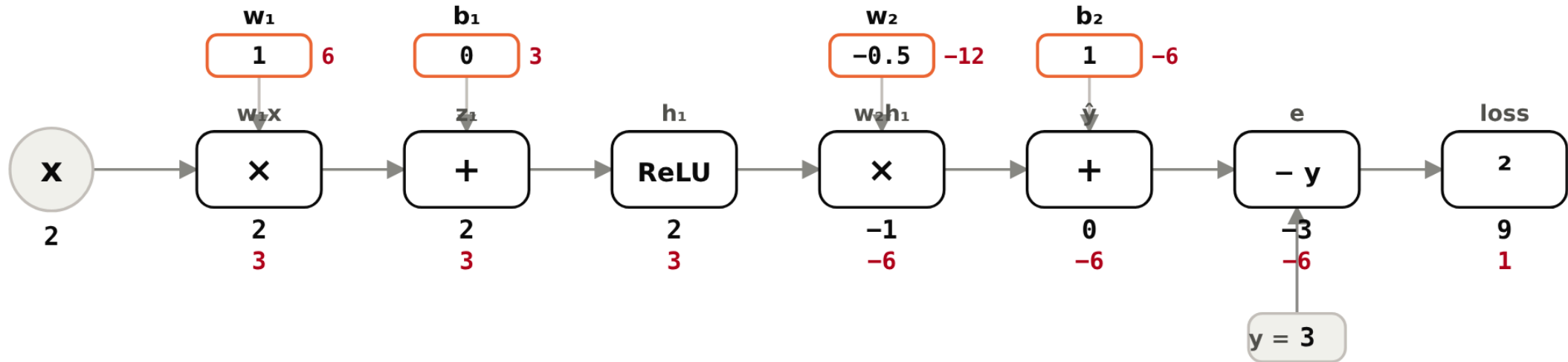
Four Rules

The slope coming from the right, g , is passed to the left

Operation	Rule	Example
$\times$	multiply g by the other input	$\hat{y} = w \cdot h$, $h = 2$, $g = -6$: w gets $2 \times (-6) = -12$
$+$	pass g unchanged	$z = w \cdot x + b$: b gets g
ReLU	pass g if $z > 0$, otherwise 0	$z = -2$: the slope stops here
2	multiply g by $2e$	$e = -4$, $g = 1$: e gets -8

One ReLU Neuron

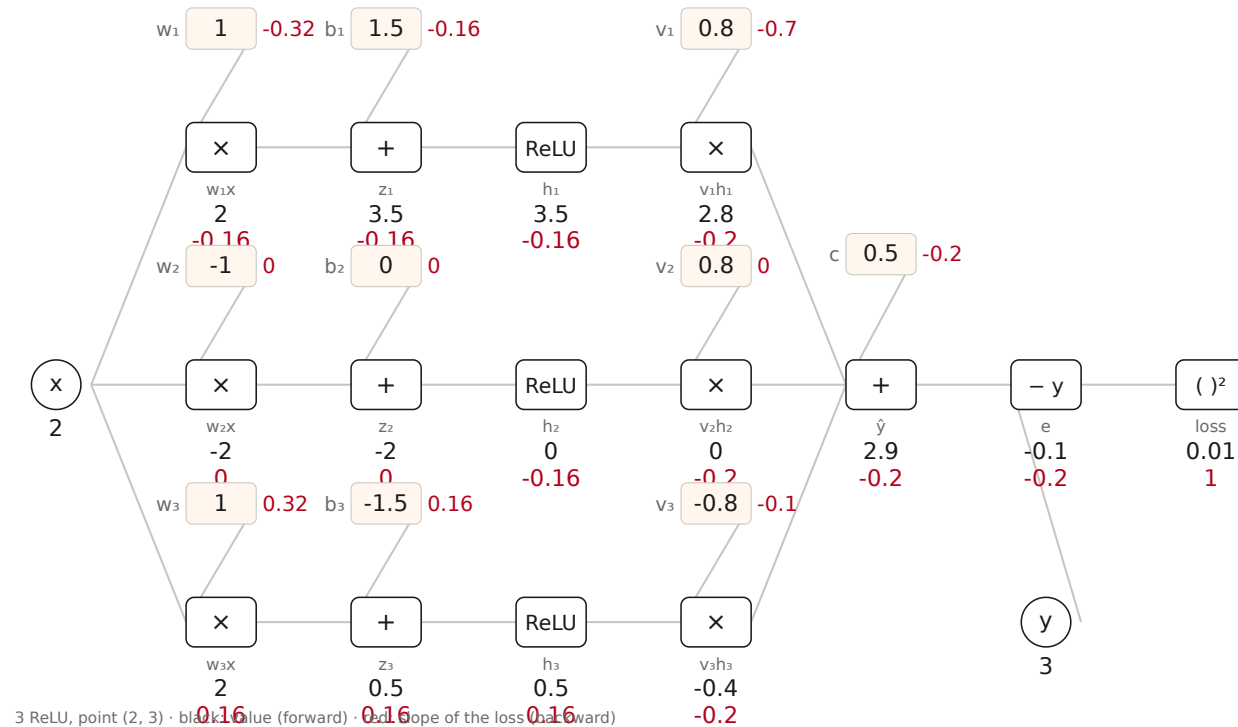
Backprop by Hand, preset "1 ReLU", point (2, 3)



- Forward: $z_1 = 2$, $h_1 = 2$, $\hat{y} = 0$, $e = -3$, $\text{loss} = 9$
- Backward: $\hat{y} -6$, $w_2 -12$, $b_2 -6$, $h_1 3$, $z_1 3$, $w_1 6$, $b_1 3$

A Closed ReLU

Backprop by Hand, preset "3 ReLU", point (2, 3)



- N2 has $z_2 = -2$, so ReLU passes 0: w_2 and b_2 get slope 0
- v_2 also gets 0, because its other input h_2 is 0

Nudge and Backprop

Backprop by Hand, “Nudge each knob”

Knob	Backprop	Nudge ($\Delta = 0.001$)
w_1	6	6.001
b_1	3	3.000
w_2	-12	-11.996
b_2	-6	-5.999

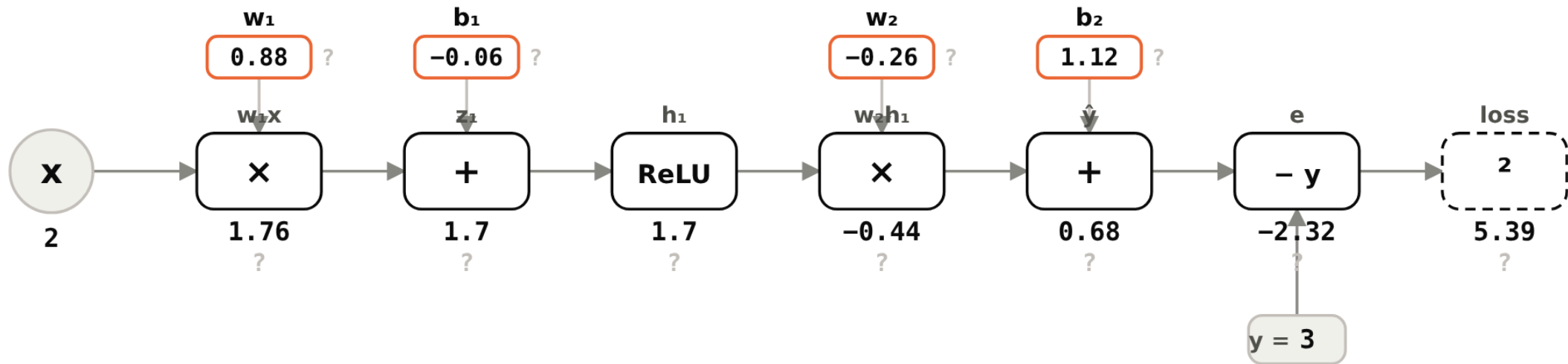
Loss computations: nudge 5 (4 knobs + 1), backprop 1 forward + 1 backward

Neuron Patch “2 layers”: 33 knobs, nudge 34 per step

- Nudge: one loss computation per knob, plus one
- Backprop: one pass forward, one pass backward, whatever the number of knobs

Take a Step

Backprop by Hand, $w \leftarrow w - lr \times \text{slope}$, lr 0.02



- Every knob moves against its slope at once: the loss for this point drops from 9 to 5.39
- Run in Neuron Patch is this step, repeated

From One Point to 120 Points

From $(\hat{y} - y)^2$ to the mean over 120 points

- Neuron Patch's loss is the mean of this one-point loss over all 120 points
- The slope of a mean is the mean of the slopes: run the same pass for each point and average

What we have

Three Parts of Machine Learning

- A model: layers that rotate, stretch, shift and fold the input space
- A loss: one number that measures how far the predictions are from the data
- A search: gradient descent, with every slope from one backward pass

Coming Up

- Week 4: working with LLMs, reading and verifying generated results
- Sep 24 (Thu): Chuseok holiday, no class

What changes when the network has 10^{11} knobs?